

Cross-Paper Tests 56–60

Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

May 15, 2026

License: Creative Commons Attribution 4.0 International (CC BY 4.0)

Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in *Inter-Self Coordination via Shared Substrate / Full Aspect Integration* (Li, April 2026), the third paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026) and *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026).

Abstract

The Paper 2 ↔ Paper 3 interface is where intra-Self governance and inter-Self governance meet. Paper 3's Full Aspect Integration (FAI) mechanism presupposes Paper 2's architectural objects — aspects, cells, DNA layers, action layers, and the instinct/reasoning separation — and extends them across the inter-Self perimeter. If this interface is not governed correctly, the two papers' commitments decouple: Paper 3 governance operates without Paper 2's structural substrate, or Paper 2 home-perimeter governance erodes under FAI participation. This note formalizes five binary operational tests — Tests 56 through 60 — that verify the Paper 2 ↔ Paper 3 interface is governed correctly. Test 56 checks lifecycle isolation during FAI events. Test 57 checks evolution isolation during FAI events. Test 58 checks that home instinct architecture is unchanged by FAI participation. Test 59 checks that vertical FAI learning provenance chains are complete and traceable. Test 60 checks Paper 2 architectural compliance as a prerequisite for FAI participation. Together, these five tests close the cross-paper adversarial gap: the shared substrate at inter-Self scope requires that the contributing entities are Paper 2-compliant Selves, and these tests verify that both the prerequisites and the interface properties hold. This note extends the test library from 55 to 60 tests.

1. Purpose and Context

The CKS theory series builds across three papers in a layered dependency structure. Paper 1 establishes the substrate/cell architecture and six foundational commitments. Paper 2 extends the architecture to aspects, Selves, and the instinct/reasoning separation within a single Self's home perimeter. Paper 3 extends coordination across the inter-Self perimeter through the shared substrate and Full Aspect Integration.

Each extension layer inherits without replacing. Paper 3 does not define new architectural objects from scratch; it extends Paper 2's objects — specifically the aspect, which becomes the unit of exchange at

inter-Self scope. FAI exchanges aspects' constituent cells, DNA-layer content, and action-layer content. The instinct layer does not exchange. The four-locus evolution-feed mechanism feeds FAI outputs back into Paper 2's home evolution machinery. Every one of these operations presupposes that a Paper 2-compliant architecture is present at each participating Self's home perimeter.

The Paper 2 ↔ Paper 3 interface is therefore both the locus of the series' most powerful capabilities and the locus of its most consequential failure modes. If home lifecycle events propagate automatically into the shared substrate, contributing entities lose governance independence. If home directed-selection updates overwrite version-fixed shared-substrate contributions, the shared substrate's provenance integrity breaks. If FAI participation modifies a contributing Self's instinct layer, the instinct/reasoning separation — Paper 2's foundational architectural spine — is violated at the inter-Self boundary. If vertical provenance chains break, FAI-derived learning cannot be traced to its source, defeating the path-retraceability commitment. And if Paper 2 compliance is absent, Paper 3 governance cannot operate at all: there are no aspects to exchange, no layers to route, no evolution machinery to feed.

Tests 56–60 operationalize these five failure modes as binary pass/fail criteria applied to governance records. They are the cross-paper interface tests for the Paper 2 → Paper 3 transition.

2. Test 56 — Lifecycle Isolation During FAI

Records to examine: Home lifecycle records during the FAI event period; shared substrate content during and after the FAI event.

Binary question: For any home lifecycle events (births, matings, deaths) that occurred during the FAI event, can records confirm these events did not automatically update the shared substrate content?

PASS indicates: Home lifecycle event records and shared substrate content are independent. Lifecycle operations within participating Selves' home perimeters — the birth of new cells or aspects, mating operations combining genetic material, deaths retiring governed entities — left the shared substrate unchanged. Any reflection of lifecycle events in the shared substrate was explicitly governance-configured, not automatic. The isolation principle is operational.

FAIL indicates: Shared substrate content changed to reflect home lifecycle events without explicit governance amendment. A new cell born in Self A during the FAI event appeared in the shared substrate without a governed contribution act; a mating operation within Self B modified a previously contributed aspect's representation in the shared substrate; a death event retired a cell whose representation persisted in the shared substrate without reconciliation. Any of these patterns indicates lifecycle isolation violation: the home perimeter and the inter-Self perimeter are not governed independently, and the shared substrate's provenance integrity is compromised.

Architectural grounding: Paper 2 governs lifecycle operations within each Self's home perimeter. Paper 3 governs what content enters the shared substrate through governed contribution acts. The two governance domains are intentionally separate. Automatic propagation from home lifecycle events to shared substrate content would collapse this separation, turning every home-perimeter lifecycle

operation into an implicit inter-Self governance act — which is not authorized under either paper's architecture.

3. Test 57 — Evolution Isolation During FAI

Records to examine: Home directed selection records during the FAI event period; shared substrate version history.

Binary question: For any home directed selection events that modified a contributed aspect's DNA during the FAI event, does the shared substrate retain the pre-directed-selection version of that aspect (not the post-update version)?

PASS indicates: Shared substrate version history shows that the contribution-time version was maintained throughout the FAI event. Home DNA updates — directed selection modifying an aspect's DNA-layer specification after the aspect was contributed — did not propagate to the shared substrate. The version contributed at FAI initiation is the version the shared substrate holds. Evolution isolation is operational, and the shared substrate can be audited against the contribution-time record.

FAIL indicates: Shared substrate version history reflects post-directed-selection home DNA. An aspect contributed at version V was subsequently updated at home to version V+1 through directed selection, and the shared substrate now shows V+1 rather than V. This constitutes evolution isolation violation: the version-fixing principle has not been applied, and the shared substrate's record no longer corresponds to what was actually contributed at FAI initiation. Participating entities may disagree about what was contributed; audit trails are unreliable.

Architectural grounding: Paper 2's evolution machinery — including directed selection — operates within a Self's home perimeter and modifies the home substrate. Paper 3's shared substrate captures aspects at contribution time and holds them under inter-Self governance. The two evolution timelines are independent by architectural design: home evolution proceeds on the home substrate; the shared substrate version is fixed at contribution. Allowing post-contribution home evolution to overwrite shared-substrate contributions would mean the shared substrate does not represent what was agreed at FAI initiation, which defeats the shared substrate's function as a governed record of inter-Self coordination.

4. Test 58 — Home Instinct Architecture Unchanged by FAI

Records to examine: Home instinct-layer records (substrate configuration, LLM configuration) pre-FAI and post-FAI; governance authority records confirming home governance authority was undiminished after FAI participation.

Binary question: After FAI participation, is the contributing Self's instinct-layer architecture — its substrate configuration and LLM configuration — identical to its pre-FAI state, with no FAI-originated changes?

PASS indicates: Instinct-layer architecture is unchanged. The contributing Self's substrate configuration and LLM configuration are identical before and after FAI participation. No FAI-originated content reached the instinct layer. Home governance authority over the instinct layer was not diminished by FAI participation. The exchange-bounding commitment — FAI exchanges reasoning-layer content, not instinct-layer content — is operational at this Self.

FAIL indicates: Instinct-layer changes coincide with FAI participation, with no other causal explanation. The contributing Self's substrate configuration or LLM configuration differs pre-FAI from post-FAI in ways attributable to FAI participation. This indicates either exchange-bounding violation — content from the shared substrate reached the instinct layer, crossing the boundary Paper 2's instinct/reasoning separation defends — or home perimeter erosion, where FAI participation modified the governance configuration of the contributing Self's instinct layer without authorized home-perimeter governance action. Both failure modes are severe: the instinct/reasoning separation is the foundational architectural spine across Papers 2 and 3, and its violation at the inter-Self perimeter would undermine the entire series' governance commitments.

Architectural grounding: Paper 2 establishes the instinct/reasoning separation: LLM weights and instinct-layer content are not subject to the coordination mechanisms that govern the reasoning layer. Paper 3 explicitly extends this boundary to the inter-Self perimeter: FAI exchange is bounded to reasoning-layer content, and instinct evolution takes no FAI input by architectural commitment. Test 58 verifies this architectural commitment is operational at the participating Self level.

5. Test 59 — Vertical FAI Learning Provenance Chain

Records to examine: Self-level directed selection events attributed to FAI learning; vertical propagation records through aspects and cells; action-layer FAI-origin records.

Binary question: For a Self-level directed selection event attributed to FAI learning, can the full provenance chain be traced backward through vertical propagation to the specific FAI event and contributing aspect?

PASS indicates: A complete provenance chain is present: the Self-level directed selection event cites a source; that source is an aspect-level proposal; the aspect-level proposal is grounded in cell-level or aspect-level action-layer records marked with FAI-origin provenance; and those records reference the specific FAI event and the aspect that was contributed during that event. Every tier in the vertical propagation is documented. The chain from inter-Self coordination event to intra-Self governance decision is unbroken and auditable.

FAIL indicates: The provenance chain breaks at one or more tiers. A Self-level directed selection event is attributed to FAI learning, but either the aspect-level proposal is undocumented, the cell-level FAI-origin record is absent, or the connection to a specific FAI event cannot be established from records. This indicates either a vertical propagation break — the evolution-feed mechanism propagated content upward without preserving provenance records at each tier — or an authorization chain gap, where a governance decision at one tier was made without traceable grounding in the tier below. Either

failure mode defeats the path-retraceability commitment Paper 1 establishes and Paper 3 extends to the inter-Self scope.

Architectural grounding: Paper 3's four-locus evolution-feed mechanism feeds FAI outputs back into Paper 2's home evolution machinery while preserving per-mechanism structure and provenance. The hand-off at FAI dissolution produces evolution-feed content that enters the home substrate with FAI-origin attribution. Vertical propagation through aspects and cells carries that attribution through the governance tiers. Test 59 verifies the full chain is intact, making FAI-derived learning auditable from the Self level down to the inter-Self event that originated it.

6. Test 60 — Paper 2 Compliance Prerequisite

Records to examine: Contributing Self's governance architecture documentation; aspects with cells, DNA layers, and action layers; at least one Self-level integration specification.

Binary question: Does the contributing Self have a Paper 2-compliant architecture — aspects organized from cells, DNA-layer and action-layer content at each tier, and at least one Self-level integration specification — as a prerequisite for FAI participation?

PASS indicates: Paper 2 architectural elements are documented and present. The contributing Self has aspects, each composed of cells. Each cell carries DNA-layer content (orchestration rules, schemas, behavioral specifications) and action-layer content (recorded task instances, outputs). The Self has at least one Self-level integration specification governing how its aspects relate and how its lifecycle and evolution operations are governed. The architecture that Paper 3's FAI mechanism presupposes is actually present.

FAIL indicates: The contributing Self lacks Paper 2-compliant architecture. Aspects are absent or undocumented, cells are not organized under aspects, DNA-layer and action-layer distinction is not maintained, or a Self-level integration specification is not present. This is not a failure of Paper 3 governance — it is a prerequisite failure. FAI cannot be conducted without the foundational architecture of the paper series in place. A Self without Paper 2 compliance has no aspects to contribute, no layers for the exchange-bounding rule to operate over, and no evolution machinery for the four-locus feed to target. Paper 3 governance cannot operate without Paper 2's structural substrate.

Architectural grounding: Test 60 is the foundational cross-paper test. It does not check a governance operation; it checks whether the governance substrate that all other tests presuppose is present. The shared substrate at inter-Self scope is a Paper 1 substrate applied at inter-Self scope; the entities participating in that substrate must be Paper 2-compliant Selves for the Paper 3 architecture to be coherent. Without Paper 2 compliance, Tests 56–59 have nothing to test: there are no lifecycle events to isolate, no version-fixed contributions to protect, no instinct layer to verify as unchanged, and no vertical provenance chain to trace.

7. The Five Tests as a Complete Interface Specification

Tests 56–60 are not five independent checks. They are a structured verification of the Paper 2 ↔ Paper 3 interface at its five critical properties.

Test 60 establishes the prerequisite: Paper 2 architecture is present. Without this, no other test is meaningful.

Tests 56 and 57 verify bidirectional isolation: home lifecycle and evolution operations do not propagate into the shared substrate, and the shared substrate does not interfere with home governance. This isolation is what allows participating Selves to maintain governance independence during FAI — a necessary property if FAI participation is to be compatible with home-perimeter sovereignty.

Test 58 verifies the instinct-layer invariant at the inter-Self boundary. The instinct/reasoning separation is the foundational architectural spine of Paper 2 and Paper 3 alike; its operational verification at the participating-Self level is what confirms the inter-Self exchange boundary is enforced, not merely asserted.

Test 59 verifies that the four-locus evolution-feed mechanism preserves provenance across the vertical tier structure. This connects the inter-Self governance events of Paper 3 to the intra-Self governance tiers of Paper 2 through an auditable chain — the property that makes FAI-derived learning governable rather than opaque.

Together, the five tests verify that the paper boundary is architecturally operational: the entities participating in Paper 3's inter-Self coordination are genuinely Paper 2-compliant Selves, and the interface between home-perimeter governance and inter-Self governance is governed with the isolation, invariance, and provenance properties the architecture commits to. Adversarial claims that the paper boundary is unclear or that Paper 3 governance can operate without Paper 2 compliance must address these specific criteria to be taken seriously.

This note extends the D5 operational test library from 55 to 60 tests.

Source Papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

How to Cite This Note

Li, W. (2026). *Cross-Paper Tests 56–60*. May 15, 2026. ORCID: 0009-0004-8065-3235.